Mistral Mind: AI-Powered Personalized Learning Platform

An AI learning assistant that adapts to each learner. It writes personalized lessons, assessments and feedback, and mixes text and images in one interface.

What it does

- Generates lessons matched to the learner's level and topic
- Creates assessments and gives feedback on answers
- Produces multimodal content (text and images)
- Offers real-time AI tutoring through a chat interface

How it works

1. Several large language models are used: **Mixtral 8x7B, LLaMA, Falcon** and **IDEFICS** (for image understanding).
2. A **model routing** step sends each request to the model best suited to it.
3. Inference runs through the **Groq API** for fast responses (under 300 ms in my tests: **[add how you measured it]**).
4. A **Gradio** interface ties it together.

Tech stack

Python, LLMs, Groq API, Gradio, Hugging Face Transformers, Generative AI, NLP

Setup

```bash
git clone https://github.com/shreyadoiphode-09/[repo-name].git
cd [repo-name]
pip install -r requirements.txt
```

Create a `.env` file in the project folder and add your own key. Never commit it:

```
GROQ_API_KEY=your_key_here
```

Run the app (replace with your own file name):

```bash
python [app_file].py
```

Results

- Response latency: **[median and slowest 5%, with your hardware and network]**

- Other results: **[add if you have them]**

Related paper

Mistral Mind: Precision AI for Personalized Learning and Development (manuscript)

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